American Computer Science League

2023-2024 ● Contest 4: Short Problem Solutions ● Elementary Division

1. Graph Theory The number of edges in a complete graph with *n* vertices is a triangular number starting with a triangle with 3 edges, then 1, 3, 6, 10, 15, 21, 28, but not counting the 8 sides of the polygon, there are 20 diagonals. This is also the formula n(n-3)/2 = 8(8-3)/2 = 20.	C. 20
2. Graph Theory Using inspection with the first letter of each city's name, the routes from Chicago to Miami are CPM, CNM, CSPM, CSHNM, CPSHNM, CNHSPM.	C. 6
3. Graph Theory Two in-between friends means paths of length 3 using 4 vertices. It is possible to send a message from Ann to each other friend except for Carol in the following ways: ADCB, ABCD, ABFE, ADEF.	B. C
4. Graph Theory Traversing a graph means using every edge without lifting your pencil. You must start and end with the 2 odd degree vertices which are Y and Z. Therefore, the only one that is a valid traversal is YZWVUXYVZ.	B. YZWVUXYVZ
5. Graph Theory By inspection there are 12 cycles in this undirected graph since they occur in pairs. They are: WZVW and WVZW, VYZV and VZYV, UVYXU and UXYVU, UVWZYXU and UXYZWVU, UVZYXU and UXYZVU, VWZYV and VYZWV.	D. 12